2/22/23

Discussing normal matrices : $AA^* = A^*A$

<u>Facts</u> : Normal $A \in M_n$

(1) Every $\lambda \in \sigma(A)$ has geometric multiplicity = algebraic multiplicity
(true for all diag/ble matrices)

(2) If $A \in M_n$ is normal, then

$$\left\{ \begin{array}{c} Ax = \lambda x \\ x \neq 0 \end{array} \right\} \iff \left\{ \begin{array}{c} x^* A = \lambda x^* \\ x \neq 0 \end{array} \right\}$$

i.e. left & right eigenvectors coincide!

(not to be confused with
$Ax = \lambda x \iff x^* A^* = \bar{\lambda} x^*$)

<u>Proof</u> Say $\boxed{Ax = \lambda x}$, $x \neq 0$, $\|x\|_2 = 1$

Construct unitary U whose first column is x and the rest are orthonormal eigenvectors corresponding to other $n-1$ eigenvalues.
(see previous theorem)

Then $A = U \Lambda U^*$, $\Lambda = \begin{bmatrix} \boxed{\lambda} & & 0 \\ & * & \\ 0 & & \ddots \; * \end{bmatrix}$

Then

$$A^* = (U\Lambda U^*)^* = U\bar{\Lambda}U^*$$

So

$$\underline{A^* x} = U\bar{\Lambda}U^* x$$

$$= U\bar{\Lambda}U^{-1}x \qquad (\text{since } Ue_1 = x)$$

$$= U\bar{\Lambda}e_1 \qquad e_1 = 1^{st} \text{ column of } n\times n \text{ identity.}$$

$$= U\begin{bmatrix}\bar{\lambda}\\0\\\vdots\\0\end{bmatrix}$$

$$= \underline{\bar{\lambda}x}$$

i.e. $A^* x = \bar{\lambda} x$

$$\Rightarrow (A^* x)^* = (\bar{\lambda} x)^*$$

$$\Rightarrow x^* A = \lambda x^*$$

(reversing $x^* A = \lambda x^* \Rightarrow Ax = \lambda x$)

(3) If x, y are (right!) eigenvectors corresponding to λ, μ, respectively, and if $\lambda \neq \mu$, then

$$x^* y = y^* x = 0$$

(i.e. every pair of eigenvectors of normal A corresponding to distinct eigenvalues are orthogonal.)

<u>Proof</u>

<u>Recall</u> something in the past:

For <u>every</u> $A \in M_n$, $Ax = \lambda x$, $y^* A = \mu y^*$, $x \neq 0$, $y \neq 0$, $\lambda \neq \mu$, imply

$$x^* y = 0$$

So for a normal matrix, since left & right coincide, (3) follows by (2).

Two important theorem

Theorem 1 Let $A \in M_n$ be normal. Then there exist orthonormal vectors $u_1, u_2, \dots, u_n \in \mathbb{C}^n$ & scalars $c_1, c_2, \dots, c_n$ such that

$$A = c_1 u_1 u_1^* + c_2 u_2 u_2^* + \dots + c_n u_n u_n^*$$

($u_j u_j^* \to$ rank 1 matrix)

In fact, $A u_j = c_j u_j$, $j = 1, 2, \dots, n$.

Proof

$$A \text{ is normal} \overset{\text{theorem}}{\Longrightarrow} \exists \text{ unitary } U = [u_1 | u_2 | \dots | u_n]$$

such that

$$U^* A U = \Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \; \lambda_n \end{bmatrix}$$

$$\Rightarrow \quad A = U \Lambda U^*$$

$$= [u_1 | u_2 | \dots | u_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} u_1^* \\ u_2^* \\ \vdots \\ u_n^* \end{bmatrix}$$

$$= \left[\lambda_1 u_1 \,|\, \lambda_2 u_2 \,|\, \cdots \,|\, \lambda_n u_n \right] \begin{bmatrix} u_1^* \\ u_2^* \\ \vdots \\ u_n^* \end{bmatrix}$$

$$= \lambda_1 u_1 u_1^* + \lambda_2 u_2 u_2^* + \cdots + \lambda_n u_n u_n^* \quad \blacksquare$$

<u>Theorem 2</u> (Spectral Theorem for hermitian matrices)

If $A \in M_n$ is hermitian ($A^* = A$), then

(a) $\sigma(A) \subseteq \mathbb{R}$

(b) $A = c_1 u_1 u_1^* + c_2 u_2 u_2^* + \cdots + c_n u_n u_n^*$

where $u_1, u_2, \ldots, u_n \in \mathbb{C}^n$ are orthonormal

& $c_1, c_2, \ldots, c_n \in \mathbb{R}$

In fact, $Au_j = c_j u_j$, $j = 1, 2, \ldots, n$

& $c_j \in \mathbb{R}$

<u>Proof</u> Out to prove (a) and then (b) follows from Theorem 1.

To prove (a)

Say $\begin{cases} A = A^* \\ Ax = \lambda x, \quad x \neq 0 \end{cases}$

Then

$$x^* A x = x^*(\lambda x) = \lambda x^* x$$

$$\Rightarrow \lambda = \frac{x^* A x}{x^* x} \quad (1)$$

Also

$$\overline{(x^* A x)} = (x^* A x)^*$$

$$= x^* A^* x$$

$$\overset{A=A^*}{=} x^* A x$$

so $x^* A x \in \mathbb{R}$ (2)

By (1) & (2)

$$\lambda = \frac{x^* A x}{x^* x} \in \mathbb{R}$$

(since $x^* x > 0$) ∎

$z = x^* A x$

$[z]^*$ 1x1 matrix

$= \overline{[z]}^T$

$= \overline{[z]} = \bar{z}$

$\boxed{\text{FYI}}$

The "spectral theorem" generalizes to arbitrary $A \in M_n$, in fact $A \in M_{m,n}$

<u>Theorem</u> For every $A \in M_{m,n}$, there exist <u>unitary</u> $U = [u_1 u_2 \dots u_m]$

$$V = [v_1 v_2 \dots v_n]$$

$$\& \quad \Sigma = \begin{bmatrix} \sigma_1 & & & & & \\ & \sigma_2 & & & 0 & \\ & & \ddots & & & \\ & & & \sigma_r & & \\ & 0 & & & 0 & \\ & & & & & \ddots \; 0 \end{bmatrix} \in M_{m,n}$$

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots \geq \sigma_r, \quad r = \text{rank}(A)$$

such that

$$\underset{m \times m}{A} = \underset{m \times m}{U} \; \underset{m \times n}{\Sigma} \; \underset{n \times n}{V^*}$$

(which can be written as

$$\boxed{A = \sigma_1 u_1 v_1^* + \sigma_2 u_2 v_2^* + \dots + \sigma_r u_r v_r^*}$$

(SINGULAR VALUE DECOMPOSITION)

2/24/23

On homework #2

Extended by a week: 3/5/23

- 2.4 P2 Typo

$$\left|\sum_{i=1}^{k} \lambda_i\right|^2 \le k \sum_{i=1}^{k} |\lambda_i|^2 = k \sum_{i=1}^{k} |t_{ii}|^2$$

similarity

algebraic

$$\le k \sum_{i,j=1}^{n} |t_{ij}|^2 = k \sum_{i,j=1}^{n} |a_{ij}|^2$$

more terms

theorem preserving Frobenius norm

- 2.4 p11 Skip (e)

$C = AB - BA$ commutator

(a) $\operatorname{trace}(C) = 0$ $\quad (\operatorname{tr}(AB) = \operatorname{tr}(BA))$

(b) specific matrices

(c) $\operatorname{rank}(C) \le 1 \xRightarrow{\text{show}} C$ nilpotent

i.e. $\sigma(C) = \{0, 0, \dots, 0\}$

- cases
- use (a)

(d) $\operatorname{rank}(C) = 0 \Rightarrow C = 0$ easy.

C $n \times n$

$\text{rank}(C) = 1$

$\dim N(C) = n-1$

$\Rightarrow$ 0 has $gm(0) = n-1$

i.e. $am(0) \geq n-1$

i.e. $\sigma(C) = \{\lambda, \underbrace{0, 0, \ldots, 0}_{n-1}\}$

But $tr(C) = 0$, $\lambda + 0 + \ldots + 0 = 0 \Rightarrow \lambda = 0$

So $\sigma(C) = \{0, 0, \ldots, 0\}$

eg $C \equiv \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

A few problems on normal matrices

(i) Let $A \in M_n$ and

$$\|Ax\|_2 = \|A^* x\|_2$$

For all $x \in \mathbb{C}^n$. Show that A is normal.

$$\|Ax\|_2^2 = \|A^* x\|_2^2$$

$$\iff (Ax)^*(Ax) = (A^*x)^*(A^*x)$$

$$\iff x^* A^* A x = x^* A A^* x$$

$$\iff x^* \underbrace{(A^*A - AA^*)} x = 0 \quad , \quad \forall x \in \mathbb{C}^n.$$

$$\overset{\text{shown}}{\iff} A^*A - AA^* = 0$$

$$\iff A^*A = AA^* \text{ , normal!}$$

(in fact, it's a characterization of matrix normality)

(2) Suppose $A \in M_n$ is normal, and

$$|\lambda| = 1$$

for all $\lambda \in \sigma(A)$. Show that A is unitary.

Since A is normal, $\exists$ unitary U such that

Theorem!
$$\begin{cases} U^* A U = D \text{ + } D \text{ is diagonal} \\ D = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) \\ \sigma(A) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\} \end{cases}$$

So
$$D^* D = \begin{bmatrix} \bar{\lambda}_1 & & 0 \\ & \bar{\lambda}_2 & \\ 0 & & \ddots \; \bar{\lambda}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \; \lambda_n \end{bmatrix}$$

$$= \begin{bmatrix} \bar{\lambda}_1 \lambda_1 & & 0 \\ & \bar{\lambda}_2 \lambda_2 & \\ 0 & & \ddots \; \bar{\lambda}_n \lambda_n \end{bmatrix}$$

$$= \begin{bmatrix} |\lambda_1|^2 & & 0 \\ & |\lambda_2|^2 & \\ 0 & & \ddots \; |\lambda_n|^2 \end{bmatrix} = I$$

Thus
$$A = U D U^*$$

&
$$A^* A = U D^* \cancel{U^* U} D U^* = U \underbrace{D^* D} U^*$$

$(U^*)^*$
$$= U U^* = I$$

i.e. A is unitary ■

(3) Say $A \in M_n$ is normal

& nilpotent.

Show $A = 0$.

(nilpotent : $\exists\, k$ such that $A^k = 0$

Equivalent $\sigma(A) = \{0, 0, \dots, 0\}$)

" $A^k = 0 \Rightarrow \lambda^k = 0 \quad \forall \lambda \in \sigma(A)$ "

<u>Normal</u> $\Rightarrow A = UDU^*$ & $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \; \lambda_n \end{bmatrix}$

$\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$

<u>nilpotency</u> $\Rightarrow \lambda_1 = \lambda_2 = \dots = \lambda_n = 0$

So $A = U \cdot 0 \cdot U^* = 0$

" Only diag/ble nilpont matrix is the zero matrix "

2/27/23

<u>Midterm</u> : Take home
Emailed: probably on Wednesday March 8.
Will include instructions.

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§ 2.6 QR-factorization

<u>Theorem</u> Let $A \in M_{n,m}$, $n \geq m$. Then there exist $Q \in M_{n,m}$ with orthonormal columns and upper triangular $R \in M_m$ such that $A = QR$.

- If $m = n$, then Q is unitary.
- If A has full rank, i.e. $\operatorname{rank}(A) = m$, then R can be chosen so that the diagonal entries of R, $r_{ii} > 0$.
- If $A \in M_{n,m}(\mathbb{R})$, then Q, R can be chosen to be real matrices.

proof

<u>Proof</u> (when A has full rank, m)

Let $A = [a_1 | a_2 | \cdots | a_m]$, i.e. the columns of A are $\{a_1, a_2, \ldots, a_m\} \subseteq \mathbb{C}^n$ and since $\text{rank}(A) = m$, they forms a linearly independent set.

Use Gram-Schmidt & normalization to produce an <u>orthonormal basis</u> for the span$\{a_1, \ldots, a_m\}$:

$$\{u_1, u_2, \ldots, u_m\}$$

Let $Q = [u_1 | u_2 | \cdots | u_m] \in M_{n,m}$

A careful analysis of Gram-Schmidt (look at its proof) reveals that in the process of creating $u_1, u_2, \ldots, u_m$, we have

$$\text{span}\{a_1, a_2, \ldots, a_k\} = \text{span}\{u_1, u_2, \ldots, u_k\}$$

$$k = 1, 2, \ldots, m$$

<u>Thus</u>

$$a_1 = r_{11} u_1 \quad \text{for some } r_{11} \in \mathbb{C}$$

$$a_2 = r_{12} u_1 + r_{22} u_2 \quad \text{"} \quad r_{12}, r_{22} \in \mathbb{C}$$

$$\vdots$$

$$a_k = r_{1k} u_1 + r_{2k} u_2 + \cdots + r_{kk} u_k$$

$$r_{1k}, \ldots, r_{kk} \in \mathbb{C}$$

Let

$$r_k := \begin{bmatrix} r_{1k} \\ \vdots \\ r_{kk} \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{C}^m, \quad k = 1, 2, \ldots, m$$

and collect them in a matrix

$$R = \left[r_1 \,|\, r_2 \,|\, \cdots \,|\, r_m \right] \in M_m$$

Notice that by construction, that

$$A = [Qr_1 \,|\, Qr_2 \,|\, \cdots \,|\, Qr_m]$$

$$= QR \quad \blacksquare$$

Notice that because of the "convenience" assumption that $\operatorname{rank}(A) = m$, we have

$$r_{ii} \neq 0$$

$$\left(\operatorname{rank}(A) = \operatorname{rank}(QR) \leq \min(\operatorname{rank}(Q), \operatorname{rank}(R)) \leq m \right.$$

$$\left. \text{so if } \operatorname{rank} A = m \Rightarrow \operatorname{rank}(R) = m \right)$$

WLOG we can assume $r_{ii} > 0$, otherwise replace u_i by $\frac{|r_{ii}|}{r_{ii}} u_i$

Therefore,

then theorem says

If $\text{rank}(A) = m$, then $A = QR$ with the diagonal entries of R being positive. In this case

$$A = QR$$

the factors Q & R are <u>unique!</u>

<u>Proof</u> (of uniqueness in this case)

Say $A = QR = Q_1 R_1$

Then

$$QR = Q_1 R_1$$

$$\Rightarrow \underbrace{Q_1^* Q}_{\text{unitary}} = \underbrace{R_1 R^{-1}}_{\text{upper triangular.}}$$

$\Rightarrow$ $Q_1^* Q$ is diagonal & diagonal entries must be equal to 1. (prove)

i.e. $Q_1^* Q = I$

i.e. $Q = Q_1$

i.e. $R = R_1$ provided that diagonal entries > 0.

Contrast with the "QR-algorithm"

(an iterative algorithm to compute eigenvalues of $A \in M_n$)

$$A = A_1 = Q_1 R_1 \qquad \text{(QR-factor)}$$

$$A_2 = R_1 Q_1 \qquad \text{(reverse factors)}$$

$$= Q_2 R_2$$

$$A_3 = R_2 Q_2$$

$$= Q_3 R_3$$

$$\vdots$$

$$A_k \xrightarrow{k \to \infty} \begin{bmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

upper triangular

$$\sigma(A) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}.$$

STORY: Accidentally discovered.